

Quantum Fluctuations and Electromagnetic Decoupling at the Event Horizon

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Abstract

This paper establishes the mathematical foundation for "quantum fluctuations" and the phenomenon of event horizon darkness. It proves mathematically that the absence of radiation is not due to a mechanical gravitational trapping of photons, but rather a direct result of Quantum Electrodynamics (QED) decoupling. The transition shifts the electromagnetic field into a pure quantum vacuum state.

1 QED Decoupling Dynamics

The Lagrangian density of Quantum Electrodynamics in curved Schwarzschild spacetime is formulated as:

$$\mathcal{L}_{QED} = -g \left(\bar{\psi}(i\gamma^\mu D_\mu - m_e)\psi - \frac{1}{4}F_{\mu\nu}F^{\mu\nu} \right) \quad (1)$$

The interaction term, which mediates the emission and scattering of light, relies entirely on the charge current $J^\mu = \bar{\psi}\gamma^\mu\psi$:

$$\mathcal{L}_{int} = -g \cdot J^\mu A_\mu \quad (2)$$

As we approach the event horizon ($r \rightarrow r_s$), the thermal evaporation mechanisms induce a critical discharge, yielding a vanishing electron density:

$$N_e(r) = N_0 \exp \left(-\lambda \frac{r_s}{r - r_s} \right) \quad (3)$$

At the precise boundary of the event horizon ($r = r_s$):

$$\lim_{r \rightarrow r_s^+} N_e(r) = 0 \implies \lim_{r \rightarrow r_s^+} \langle J^\mu \rangle = 0 \quad (4)$$

Consequently, the interaction Lagrangian vanishes completely ($\lim_{r \rightarrow r_s^+} \mathcal{L}_{int} = 0$), mathematically proving that the electromagnetic field completely decouples from matter.

2 Quantum Fluctuations and Vacuum Transition

Due to the absolute decoupling ($J^\mu \rightarrow 0$), the classical expectation value of the field vanishes ($\langle F_{\mu\nu} \rangle = 0$). The energy-momentum tensor undergoes a fundamental **quantum fluctuation (transition)** into a pure vacuum expectation value:

$$\lim_{r \rightarrow r_s^+} \langle T_{\mu\nu} \rangle = \langle 0 | \hat{T}_{\mu\nu}^{(vacuum)} | 0 \rangle_{Renorm} \quad (5)$$

Evaluating this using the Boulware Vacuum State at the horizon demonstrates that the temporal component records pure vacuum stress, generating Hawking evaporative radiation rather than trapped classical light:

$$\langle \hat{T}_t^t \rangle_{vacuum} \sim -\frac{\hbar c}{768\pi^2 r_s^4 \left(1 - \frac{r_s}{r}\right)} \quad (6)$$

3 Observed Intensity and Conclusion

Integrating the geometrical redshift with the quantum decay function of electrons yields the total optical intensity I_{obs} :

$$I_{obs}(r) = I_0 \cdot \left(1 - \frac{r_s}{r}\right)^2 \cdot \frac{N_e(r)}{N_0} \quad (7)$$

$$\lim_{r \rightarrow r_s^+} I_{obs}(r) = I_0 \cdot (0) \cdot (0) = 0 \quad (8)$$

This confirms that the darkness of a black hole is a twofold phenomenon: a geometric infinite redshift and a **quantum zero** induced by the complete absence of interacting charged matter ($N_e = 0$). The "light trapping" concept is therefore a classical illusion superseded by QED vacuum fluctuations.